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Wednesday LEOP Simulation — Instructor Pack

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Classification: Internal

Wednesday LEOP Simulation — Instructor Pack

Confidential — instructor and simulator operator only.

This pack covers the Wednesday simulation day. Two student teams swap operator/observer roles between two passes. Anomaly injection choices are listed; the instructor picks per team strength after Tuesday's training quiz.

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1. Day at a glance

When	What	Mode
09:00 – 09:30	Standup, scenario brief to both teams	Instructor-led
09:30 – 11:30	**Pass 1** — Team A operates; Team B observes	LEOP sim
11:30 – 12:30	Lunch + observer prep	—
12:30 – 12:45	Reset + brief for Pass 2	Instructor-led
12:45 – 14:45	**Pass 2** — Team B operates; Team A observes	LEOP sim
14:45 – 16:00	Joint debrief, observer presentations	Plenary
16:00 – 17:00	Anomaly-ticket completion + shift logs	Individual

Each pass is 90 min real time / 90 min sim time (1× speed).

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2. Scenario configuration

- **Breakpoint:** BP-2 — LEOP Day Start (existing scenario `scenarios/first_contact.yaml`).
- **Phase at start:** 3 (BOOTLOADER_OPS).
- **Initial visibility:** Iqaluit, AOS at scenario T+0.
- `override_passes` flag set so contact remains for the pass duration.
- **Speed multiplier:** 1× (real time).

[Claude Code: confirm the scenario CLI invocation. Expected form: `sim run scenarios/first_contact.yaml --duration 5400 --inject <anomaly_yaml>`]

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3. Procedure expectations

Operator team is expected to drive the spacecraft from phase 3 (bootloader, no antennas) to phase 4 (LEOP, full HK, detumble complete) in roughly that order:

1. **LEOP-001** First Acquisition of Signal & OBC Boot
2. **LEOP-006** Time Synchronisation
3. **(LEOP-002)** Initial Health Check

4. **LEOP-007** Sequential Power-On (this is the long one — battery heater, reaction wheels, AOCS detumble; see procedure for full ordering note)
5. (Optional, if time permits) **LEOP-005** Sun Acquisition

Pass 1 typical end-state: phase 4, antennas deployed, app SW running, htr_bat ON, aocs_wheels ON, body rates < 0.05 °/s.

Pass 2 typical end-state: same, plus payload-bus on standby, FDIR rule sets verified, ready for commissioning Thursday.

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4. Anomaly injection menu

Pick **one** primary injection per team and one optional secondary ("rolling injection") if the primary is resolved cleanly. Files named are scenario YAML files in `~/SpaceMissionSimulation/configs/eosat1/scenarios/`; trigger them through the scenario operator UI mid-run.

Severity	Scenario file	What students see	Recovery procedure
Easy	`ttc_no_tm_at_aos.yaml`	Beacon present, full HK never reports despite OBC_BOOT_APP	CTG-019 No TM at Pass Start
Easy	`eps_overcurrent.yaml`	Sudden overcurrent on a power line during sequential power-on	CTG-012 Overcurrent Response
Medium	`aocs_actuator_stuck.yaml`	One reaction wheel locked at 0 RPM during detumble	CTG-007 Reaction Wheel Anomaly
Medium	`aocs_star_tracker_failure.yaml`	Primary star tracker degrades at end of detumble	CTG-008 Star Tracker Failure
Hard	`obc_watchdog.yaml`	OBC watchdog reset mid-deploy	CTG-010 OBDH Watchdog Recovery
Hard	`obc_bus_failure.yaml`	CAN bus A drops mid-sequential-power-on	CTG-017 Bus Failure Switchover
Very hard	`gs_antenna_failure.yaml`	Iqaluit antenna degraded mid-pass; team must defer rest of pass to Troll	CTG-020 Ground Station Antenna Failure

[Claude Code: cross-check filenames against simulator. The four *medium* and *hard* scenarios are listed in `scenarios/aocs_*.yaml`, `obc_*.yaml`, etc.]

Injection cue card

For each injection, the instructor needs to know:

1. **Cue moment** — at what point in the LEOP procedure to inject.
2. **First indicator** — what the operators will see first (telemetry parameter, FDIR alarm, voice cue).
3. **Pass criterion** — what "successful recovery" looks like for the pass.

4. **Five debrief questions** to use after the pass.

[Claude Code: produce a one-page cue card per scenario.]

5. Pass operator-side criteria

End-of-pass GO/NO-GO criteria (instructor uses for assessment):

Criterion	Pass 1 (Team A)	Pass 2 (Team B)
OBC application SW running	✓	✓
Onboard time set, drift < 0.5 s	✓	✓
Antennas deployed, high-rate link	✓	✓
Battery heater ON	✓	✓
Reaction wheels ON, AOCS in DETUMBLE	✓	✓
Body rates < 0.05 °/s	optional	✓
Anomaly ticket filed for any injection	✓	✓
FDIR rule sets enabled and verified	—	✓

Each criterion contributes to the team's **simulation performance** mark (Week 3 assessment row).

6. Coaching during the shift

The instructor (acting as Mission Director) sits behind the FD seat and coaches by exception only:

- **Voice-loop discipline.** If the team starts cross-talking, stop, name the rule, restart.
- **Procedure compliance.** If a position commands without FD GO, stop, log it, treat as a safety violation.
- **Anomaly response.** If the team is stuck > 5 minutes, drop a hint at the FDIR-panel level (don't name the procedure).
- **Time pressure.** Maintain the real-time clock — no pausing the sim for discussion. Pass closes when the orbit closes.

7. Debrief (joint)

Structure of the 75-minute debrief at end of day:

1. **Hot-wash (15 min).** Each team's FD reads their shift log highlights. No critique yet.
2. **Anomaly walk-through (20 min).** Operator team walks through the anomaly response; observer team challenges.
3. **Peer-review worksheet readout (20 min).** Each observer team reads its four-axis worksheet aloud (team / comms / space / technical). Operator team responds with one accept and one defend per axis.

4. **Lessons-learned synthesis (15 min).** Instructor consolidates into a single list. Logged in the cohort lessons-learned register.
5. **Brief Thursday (5 min).** Reading task: at minimum the COM-001 through COM-008 procedure cards before tomorrow.

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8. Safety and red lines

- **No commanding without FD authorisation.** Violation → instructor calls the shift, team logs the violation, no penalty if learnt-from.
- **No simulator-pause to win an exercise.** The clock runs.
- **No silent FD.** The Flight Director must be on the loop. If the FD is over-driven, the instructor calls a 30-second hold and resets the position.

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9. References & files

- Simulator scenario: `configs/eosat1/scenarios/first_contact.yaml`
- Anomaly scenarios: see §4 table
- Procedure files: `configs/eosat1/procedures/leop/*.md`
- MCS positions: `configs/eosat1/mcs/positions.yaml`
- Telemetry limits: `configs/eosat1/mcs/limits.yaml`
- ECSS-E-ST-70-11C — Space segment operability
- NASA *Mission Operations Handbook* — https://nodis3.gsfc.nasa.gov/
- ESA *Mission Operations Concept* — https://www.esa.int/Enabling_Support/Operations